

AI readiness of the federal statistical system: host-level findings

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Draft. 16 recognized bodies and three reference hosts, one cycle, one client identity. Written from `cc_tasks/2026-09-09_report_draft.md` against the graph as it stood on 2026-09-09. Every number below is a registered Result quoted by name; nothing is typed into the prose.

What was measured, and over what

The frame is the recognized agencies and units of the federal statistical system: 16 bodies. They are taken from the Interagency Council on Statistical Policy's own charter, which enumerates its membership under an explicit flag legend rather than in prose. Recognition is a legal status conferred under CIPSEA 2018 (44 U.S.C. 3561, 3562), so the roster is a parse of an authoritative list and not a judgement about who counts. The charter and the Council's public About page disagree about 1 body, the Social Security Administration's research office. The charter marks it recognized and the About page does not, and the charter was named the authority before either document was read.

Three further hosts appear in the measurement and in none of its rates: `data.gov`, NIST and GSA. These are reference hosts. A federal catalog and two data portals are the nearest available control for what a machine-facing government site looks like when someone has tried. They are judged on the host-level checks only and enter no agency denominator, because placing a catalog beside a statistical agency above that level compares two different kinds of thing.

Counting surfaces rather than bodies, the target list carries 65 declared surfaces across 22 netlocs. Netlocs exceed bodies because each reference host declares a machine entry point on its own hostname. Every surface is declared, by the roster or by the operator, and no rule selects one. Three selection rules were built for this frame and all three failed; the record of why is in the design decisions.

The client, and what it did

One identified client, one request per second per host, `robots.txt` obeyed, no forms, no logins, no query-string fuzzing. The user agent names the project and links to its source. The scanner has never retried a host under another identity, because a refusal is itself a measurement and a disguise would destroy it.

The cycle issued 2341 requests across 24 netlocs, which is more netlocs than the target list names. The excess is not an accident and is discussed under what the matrix cannot see. It produced 2429 observations and 634 findings, and every finding re-derives byte for byte from its stored observations.

The six checks

Each asks something a machine would need before it could use a body's statistics without a person in the loop.

A4 asks whether the host serves a `robots.txt` and whether it permits an identified, compliant client to read the page in question. A site with no such file has not refused; it has said nothing, and a client has to guess.

A5 asks whether the host offers a way to discover what it publishes, through a sitemap it declares itself or one of the newer machine-directed files. Discovery is the difference between a catalogue and a maze.

A10 asks whether a deep link behaves. A page that answers a nonsense URL with a cheerful page and a success status tells a machine that everything exists, which is worse than an honest absence.

A11-declared asks whether the host declares a machine layer anywhere in its own markup: an API, a bulk endpoint, a data catalogue.

A12 asks whether what the host declares and what it enforces agree, comparing the permission stated in `robots.txt` against the answer a compliant client actually receives on the same path. It is a candidate check, adopted by nobody, and it enters no fraction here.

G1-D asks whether published figures carry the fields that make them interpretable: a measure of uncertainty, a suppression flag, a reliability marker. An estimate without them can be read by a machine and cannot be used responsibly by one.

The matrix

One row per body, one column per check, one cell per verdict. `pass` and `fail` are measurements. `error` means this scanner could not observe the surface, which is a fact about the scanner and its reception, never a fact about the product. The last column counts how many of the probes issued against the body's own page the host answered with a refusal status.

The five surface-judged checks are read from each body's home page, and the coherence check from its well-known set. Which surface a cell was measured on is recorded on every row of the machine-readable copy beside this one. It matters: a `robots.txt` that permits the front door can disallow a particular product, so a flagship page and a home page need not receive the same answer. This cycle found that disagreement on 15 cells across 6 bodies, and a single combined cell would have had to pick one without saying which.

Agency	A4	A5	A10	A11-declared	A12	G1-D	Refused of probed
BEA	pass	fail	pass	pass	pass	fail	0 of 70
BJS	pass	fail	pass	pass	pass	fail	0 of 59
BLS	error	error	error	error	fail	error	33 of 35
BTS	error	error	error	error	fail	error	32 of 33
CENSUS	pass	pass	pass	pass	pass	fail	4 of 60
DRSMSU	fail	fail	pass	fail	fail	fail	0 of 67
EIA	pass	pass	pass	pass	pass	fail	1 of 65
ERS	pass	fail	pass	pass	pass	fail	0 of 60
NAHMSAPHIS	pass	fail	pass	pass	pass	fail	0 of 58
NASS	pass	fail	pass	pass	pass	fail	0 of 67
NCES	error	error	error	error	error	error	0 of 60
NCHS	pass	fail	fail	fail	pass	fail	0 of 34
NCSES	pass	pass	pass	pass	pass	fail	0 of 65
ORES	error	error	error	error	fail	error	32 of 33
SAMHSACBHS	pass	fail	pass	pass	pass	fail	0 of 62
SOI	pass	fail	pass	pass	pass	fail	0 of 62
<i>this report</i>	-	-	-	-	-	-	planned host: GitHub Pages under this repository; to be measured on publication

Reading it by column

Serving robots.txt. 11 bodies of 12 observable ones serve a `robots.txt` that permits this client to read the page. The single failure serves no such file at all: not a refusal, an absence, and a machine meeting it has to assume rather than read. 4 bodies could not be observed on this check, and the same 4 recur in every column below.

Discovery. 3 of 12 offer a discoverable index of what they publish; 9 do not. Discovery is the weakest host-level result by a wide margin, and the consequence is direct: a machine that cannot enumerate what a site publishes cannot tell what it has missed. Every sitemap any host declared was followed, including ones declared on a neighbouring hostname, so none of these failures is an artefact of the scanner declining to look.

Deep links. 11 of 12 answer a deliberately invalid URL honestly. This is the best result on the page and it deserves less credit than it looks: it measures the absence of a specific pathology rather than the presence of a capability.

Declared machine layer. 10 of 12 declare somewhere in their own markup that a machine reader is expected. Declaring is not providing, and this check does not follow the declaration to see whether anything answers at the other end.

Declared against enforced. The candidate check finds 11 bodies coherent and 4 incoherent, of 15. An incoherent host publishes a `robots.txt` granting access and then declines to serve the client that obeys it. Three of them are the bodies discussed in the next section, and the fourth publishes no `robots.txt` at all.

Uncertainty fields. 0 bodies of 12 expose the fields that make an estimate interpretable at the host level. The upper bound of the ninety-five percent interval on that rate is 0.242494 at this denominator, which is the honest way to say that a zero here is not proof of universal absence. It is measured on the body's own front page and not on a data product, which flatters nobody and is a limit of the host-level view rather than a finding about statistical practice.

The reference hosts

Agency	A4	A5	A10	A11-declared	A12	G1-D	Refused of probed
GSA	pass	fail	pass	pass	pass	fail	0 of 11
NIST	pass	fail	pass	pass	pass	fail	0 of 11
data.gov	pass	fail	pass	pass	pass	fail	0 of 12

All 3 reference hosts serve `robots.txt`, declare a machine layer, answer deep links honestly and are coherent between declaration and enforcement. All three fail discovery. All three also fail the uncertainty check, which for a catalogue is the expected and uninteresting answer: it carries no estimates, so it has no uncertainty fields to expose. They appear here for contrast and in no rate above.

Three bodies will not serve a compliant machine

Three bodies decline to answer a client that identifies itself, says where to complain about it, asks for no more than one page per second, and obeys the `robots.txt` those same hosts publish. They are the Bureau of Labor Statistics, the Bureau of Transportation Statistics and the Social Security Administration's research office. Each returns a refusal status on effectively every request. Their rows above read error throughout, which is the correct reading: this instrument did not find those sites wanting, it was not allowed to look.

The behaviour is neither a transient nor a sampling accident. It has now been recorded in 5 separate measurements, taken on three different days under two different user-agent strings. The count survives a change in how this project names errors: the earliest filed the refusal under a general client-error class, because the closed set of names had no member for a refusal until later. The number of bodies refusing has not moved, standing at 3 on the first look and the same on the most recent.

Two things follow, and only two. First, the coherence check fails on all three: each publishes a `robots.txt` that grants access and then refuses the client that honours it. Whatever the intent, the machine-readable statement and the machine-observable behaviour disagree, and a client has no way to discover which one is real except by being turned away. Second, no rate in this report describes them, and none can. They are in the frame, they are counted in the denominator of nothing, and the space they occupy in the matrix is the shape of what is not known.

What does not follow is any account of why. This scanner sees a status code. Bot management, a content delivery configuration, a deliberate policy and an unnoticed default all look identical from outside, and the only honest thing to report is the behaviour and its persistence. Distinguishing them needs either

the operator's own logs or a request made from a different vantage point, and both are named as future research rather than guessed at here.

A fourth body could not be observed either, for an unrelated reason: its host timed out or closed the connection on most probes. That is this scanner's problem or the network's, and it is recorded as error for the same reason the refusals are.

What the products offer a machine: partial

Everything above measures a host. This section measures products, and it covers less than half the frame. Only 9 of 16 bodies have had a flagship product declared for them, and the remaining 7 carry a host row and nothing else. Those bodies appear below marked as undeclared. They are not failures. A product check cannot be asked of a product nobody has named, and quietly dropping them would shrink the frame without saying so.

Across the 16 declared surfaces, 8 of the 10 product checks return not a single pass. None of these is present on any declared flagship surface measured in this cycle: structured markup describing the data, a machine-readable licence, a declared vintage, a working pointer to the current release, a changelog, a catalogue entry, or an API a client could call. Two checks are not at zero. A bulk download in a non-proprietary format is offered by 3 of 16 surfaces, and 3 serve their substantive content without requiring a browser to execute code.

A zero at this denominator is not proof of universal absence, and the report will not let it be read as one. With 0 passes in 16 surfaces, the upper bound of the ninety-five percent score interval is 0.193608, which is the number a reader should carry rather than the zero. Hanley and Lippman-Hand give the same magnitude by their rule of thumb for zero events. The bound quoted here is the score interval, computed once and registered rather than worked out in a sentence. The correct statement is that the true rate is unlikely to exceed that bound, not that it is nothing.

The interval is wide because the population is small, and the population is small because most bodies have not declared a product. That is the single cheapest improvement available to this measurement, and it costs a list rather than a cycle.

What the matrix cannot see

Pages as a browser renders them. Every check here reads what the server sends. A site that assembles its content in the reader's browser looks empty to this scanner, and to any client that does not run code. That is a real finding about machine access and a poor description of what a person sees.

Whether any machine actually uses the data. Nothing here observes use. Access and uptake are different questions, and only the first is measured.

Anything about the three bodies that refused. Their cells are error and will stay that way under this method. No rate in this report includes them, and no reader should infer that their absence from a numerator means anything about their products.

Most of the frame, at product level. The product section covers 9 bodies of 16. Every product statement in this report is about those, and the wide intervals say so.

The scanner's own reach. The cycle contacted more netlocs than the target list names, which bounds what "the frame" means. Two bodies declare their sitemap on a neighbouring hostname and the scanner followed the declaration, which is what the discovery check is for. A closed contact list and a discovery check that reads what the host actually says cannot both hold; this cycle took the second and is reporting the choice rather than having made it in advance. One consequence is on the log: those two requests went out without first reading the neighbouring host's own robots.txt, which this scanner's manners require. That is a defect in the collector rather than a finding about anyone, and it is queued.

The same declarations could have been declined instead, so the count of discovery failures attributable to a declaration the scanner refused to follow is registered at 0 for this cycle. That is zero, and it is registered rather than omitted: a caveat that is absent and a caveat measured to be empty read alike in prose and are not the same claim.

One failure this instrument cannot yet name. Exactly 1 observation in the cycle carries an error class of unknown: the map of failures does not have a name for what happened. It is counted and reported rather than folded into a neighbouring class, because an aggregate is only worth having if nothing was quietly swept into it.

What moved since the previous cycle, and what moved it

The instrument moved, and the frame moved with it. Four checks changed rule between the two cycles, and the frame went from thirteen bodies plus a foreign statistical office to 16 plus three reference hosts. A reader who compares a rate across the two is comparing two instruments over two populations.

different frame: cycle 2 measured 13 agencies + StatCan, cycle 3 measures 16 + 3 reference hosts

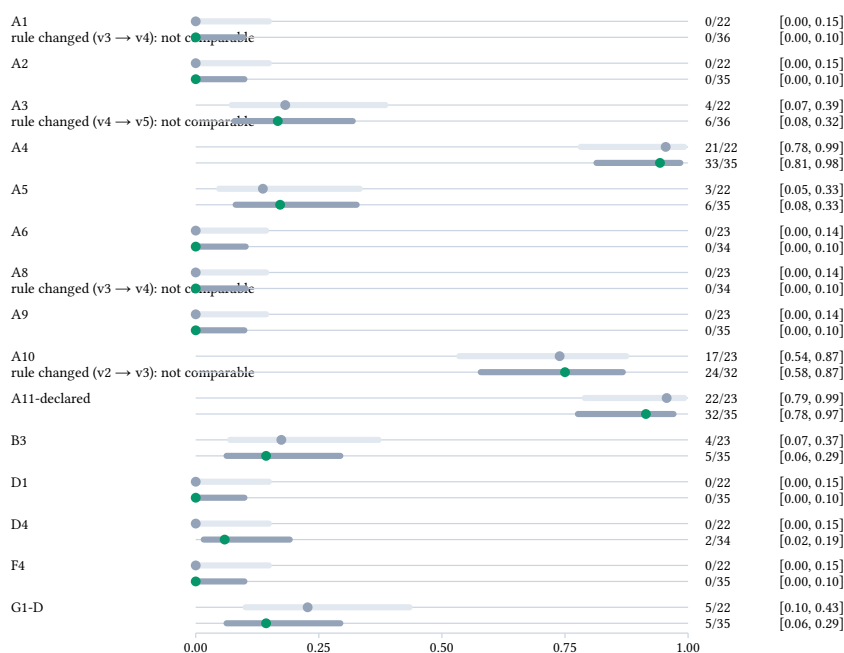


Figure 1: Pass rate per check, previous cycle beside this one

The figure marks every check whose rule changed as not comparable, and it draws no line and no arrow between the two points. A difference between two measurements is not a direction of travel, and two cycles taken a day apart on federal publication schedules are not a rate of change.

The rule changes were all instances of one defect, found three times and fixed three times: a check was reaching a verdict about a probe nobody had observed. A connection closed mid-request was being read as a product failing rather than as the scanner failing to see. Each fix turns those cases into error, which is why the error counts in this report are larger than a naive reading of a previous cycle would suggest, and why they should be. Each change was demonstrated against local fixtures before the cycle ran. Those fixtures' expected verdicts are now derived from what each collector actually dispatches, rather than written by hand, because a hand-written expectation turned out once to be wrong.

What this cannot answer, and what would

Why three bodies refuse an identified client. A status code cannot distinguish a policy from a default. Asking the operators, or reading their own access logs, is the only method that separates them, and it is the highest-value open question in this report.

Whether a third party sees what we see. Every observation here comes from one vantage point under one identity. A request from elsewhere, or under a common crawler's name, would establish whether the refusals are about this client or about machines in general.

What the products offer, for the rest of the frame. The product matrix needs a declared flagship for the remaining bodies. That list now exists as docs/design/fss_flagship_declarations.md; it is dated after the cycle this report describes and changes nothing here, because a report measures what was declared when it ran. Until a later cycle runs under that list, the product findings describe a minority of the system.

Whether the contact bound should be closed or open. Following a sitemap declared on a neighbouring host is either correct discovery or scope creep, and this cycle did it without having decided. One resolution bounds such following to the frame and records an off-frame declaration as an

observation in its own right. The other states the bound as the frame plus whatever the frame's own hosts declare.

This report, measured by its own checks

The last row of the matrix is this report's own host, and it carries no verdicts. The planned host is GitHub Pages under this repository, to be measured on publication; it is not scored, because the report has no host yet. It exists as a file in a repository, so it serves no `robots.txt`, declares no machine layer and answers no deep links. Scoring it now would mean inventing answers for a site that does not exist.

What the design calls for once it is published is on the record and is not vague. It wants a machine-readable copy of the matrix beside the readable one, and both already exist as the comma-separated and structured files accompanying this draft. It also wants a declared licence, a catalogue entry, markup identifying the matrix as a dataset, and a `robots.txt` permitting an identified client to read all of it. The instrument is meant to be turned on its own output. Until it can be, this section says so rather than reporting a row nobody measured.

Method appendix

Where the numbers live. Every value in this report is a registered Result in the project's artifact graph, quoted by name and resolved at build time. Nothing was typed. The build refuses to write this file if a single reference fails to resolve, and a lint over the built text refuses any numeral in prose that did not come through a reference. The matrices are rendered from the cycle's own data rather than transcribed. Each cell in the machine-readable copies names the Finding identity it came from, so any cell can be walked back to its observations, and those to the retained response bodies.

Client identity. `ai-readiness-kg-scanner/0.2` (`+https://github.com/brockwebb/ai-readiness-kg`). One identity, declared in parameters and read from there by everything that reports it. No host was ever retried under another.

Manners. RFC 9309 obeyed: `robots.txt` read before fetching, disallowed paths not fetched and recorded as a decision rather than as a failure. One request per second per host, one worker, no forms, no logins, no query-string fuzzing. Retries only on 429 and 503.

Verdicts. `pass` and `fail` are measurements. `error` means the collector could not observe and is excluded from every denominator; it is never a product failure. `n.a.` means there was nothing of that kind to check. `not declared` means no product has been named.

Intervals. Wilson score intervals, computed by the project's own rollup module and registered per check. Wilson (1927) for the interval. Brown, Cai and DasGupta (2001) and Newcombe (1998) for why a score interval rather than a normal approximation at proportions near zero or one. Hanley and Lippman-Hand (1983) for the rule of three at zero events.

Controls. Six local fixture servers are scanned in every cycle alongside the real hosts. One passes every check and one fails every check. One refuses an identified client and one resets the connection. One answers normally but kills the invalid-route probe, and one answers every GET while resetting every HEAD. Their expected verdicts are derived from what each collector dispatches, not written by hand: a hand-written expectation was wrong once.

Rules that judged this cycle.

Check	Rule that judged this cycle
A1	RULE-A1-v4
A10	RULE-A10-v3
A11-declared	RULE-A11-declared-v2
A12	RULE-A12-v1
A2	RULE-A2-v3
A3	RULE-A3-v5
A4	RULE-A4-v1
A5	RULE-A5-v1
A6	RULE-A6-v2
A8	RULE-A8-v4
A9	RULE-A9-v1
B3	RULE-B3-v2
D1	RULE-D1-v3
D4	RULE-D4-v2
F4	RULE-F4-v3
G1-D	RULE-G1-D-v1

Requests issued, per netloc.

Netloc	Requests
bjs.ojp.gov	183
catalog.data.gov	13
data.gov	1
data.nist.gov	13
nces.ed.gov	62
ncses.nsf.gov	184
open.gsa.gov	13
samhsa.gov	1
www.aphis.usda.gov	62
www.bea.gov	247
www.bls.gov	38
www.bts.gov	37
www.cdc.gov	219
www.census.gov	246
www.data.gov	15
www.eia.gov	143
www.ers.usda.gov	243
www.federalreserve.gov	63
www.gsa.gov	15
www.irs.gov	181
www.nass.usda.gov	247
www.nist.gov	15
www.samhsa.gov	63
www.ssa.gov	37
total	2341

Files beside this report. scan_matrix_tierA_2026-09-09.csv and .json, the host-level matrix; scan_matrix_tierC_2026-09-09.*, the reference hosts; scan_matrix_product_2026-09-09.*, the partial product matrix. Every row carries its Finding identities.

Provenance. Cycle scan_2026-09-09, parameter hash 7ee55f512e8473e6d74ab8c0d089d694078baa8548b60f2b5f22fed66728d6a6. The event log is the source of truth; the graph and the matrices are projections of it and are rebuilt by replay. Design decisions DD-059 (the frame and the tier separation), DD-060 (one client identity) and DD-061 (control tables derived from collector dispatch) govern what this report may say.